Hao Zhang

Ph.D. Candidate & Engineer

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Gender: Male
Date of Birth: Nov. 5, 1994

Research Interests

Intelligent Ships, Ship Weather Routeing Optimization, Tropical Cyclone Avoidance Technology, Ship Sailing Risk Assessment, Ship Speed Loss Prediction, Marine Meteorological Services.

Education

2024.09–Present **Ph.D. in Traffic Information Engineering and Control**, *Shanghai Maritime University*, Shanghai
Research on ship weather routeing and navigation safety.

2017.09–2020.07 **M.Eng. in Traffic Information Engineering and Control**, *Shanghai Maritime University*, Shanghai
Focus on marine sensor networks and target localization.

2013.09–2017.07 **B.Eng. in Navigation Technology**, *Shanghai Maritime University*, Shanghai
Focused on ship navigation and safety.

Work Experience

Leading the operational ship weather routeing services and R&D for ocean-going vessels.

2025.10–Present **Engineer**, *Third Division, Department of Special Support*, China Meteorological Administration (CMA)

2020.06–Present **Deputy Chief**, *Ship Weather Routeing Section*, CMA Meteorological Routeing Centre

2022.03–2024.06 **Engineer**, *CMA Meteorological Routeing Centre*

2020.07–2022.03 **Assistant Engineer**, *CMA Meteorological Routeing Centre*

Research Projects

As Principal Investigator (PI)

2025.10–Present **PI**, *High-Quality Development Project*, CMA Meteorological Routeing Centre
Arctic navigation risk forecasting and ship route planning technology.

2024.06–2025.06 **PI**, *High-Quality Development Project*, CMA Meteorological Routeing Centre
Ship speed loss prediction based on marine meteorological big data.

2022.01– 2023.01 **PI**, *Youth Fund Project*, CMA Meteorological Routeing Centre
Development of optimal route sets based on multi-objective genetic evolutionary algorithms.

As Major Contributor

2024.01– Present **Core Member**, *National Key R&D Program*, China
Key technology development for ship weather routeing (2023YFC3107900).

2023.01– 2025.12 **Core Member**, *National Key R&D Program*, China
Integration and demonstration research of marine meteorological products (2022YFC3004200).

2022.05– 2025.12 **Core Member**, *China Marine Engineering*, CMA
System construction for national-level ship weather routeing sub-systems.

2021.05– 2024.05 **Researcher**, *National Key R&D Program*, China
Real-time monitoring and autonomous information service for marine meteorology (2019YFC1510105).

Selected Publications

2025 **Zhang, H.**, Niu, G., Liu, T., et al. Status Quo and Future Prospects of China's Weather Routing Services for Ocean-Going Business Vessels. *Oceans*, 6(3). DOI: 10.3390/oceans6030038.

2023 **Zhang, H.**, Rao, Y., Liu, T., et al. Optimizing the routing of ship's tropical cyclone avoidance: a case study of typhoon "Doksuri". *Journal of Marine Meteorology*, 43(4):1-10.

2019 Mei, X., Wu, H., Xian, J., Chen, B., **Zhang, H.**, and Liu, X. A Robust, Non-Cooperative Localization Algorithm in Ocean Sensor Networks. *Sensors*, 19(12), 2708.

Skills & Teams

Programming Python, PostgreSQL

Teams Key Innovation Team of CMA - Ship Weather Routeing Team (2023–Present).